

CN Unit 3 Question Bank — Network Layer & Routing

Computer Networks (CN) · Unit 3 · Network Layer & Routing · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. Explain IPv4 Addressing with a neat diagram and a suitable example.
2. Explain Subnetting with a neat diagram and a suitable example.
3. What is meant by Distance Vector Routing? Discuss its advantages and limitations.
4. Discuss the role of Link State Routing in Network Layer & Routing.
5. What is meant by IPv6? Discuss its advantages and limitations.
6. Discuss the role of ICMP, ARP and NAT in Network Layer & Routing.

Part B — Long answer (10 marks)

1. Explain IPv4 Addressing with a neat diagram and a suitable example.
2. Explain Subnetting with a neat diagram and a suitable example.
3. Explain Distance Vector Routing with a neat diagram and a suitable example.
4. Illustrate Link State Routing with a worked example.
5. Explain IPv6 with a neat diagram and a suitable example.
6. Compare ICMP, ARP and NAT with Distance Vector Routing. When would you prefer one over the other?
7. Explain IPv4 Addressing with a neat diagram and a suitable example.
8. Explain Subnetting with a neat diagram and a suitable example.
9. Explain Distance Vector Routing with a neat diagram and a suitable example.
10. Discuss the role of Link State Routing in Network Layer & Routing.
11. Illustrate IPv6 with a worked example.
12. Explain ICMP, ARP and NAT with a neat diagram and a suitable example.